

MaskGWM: A Generalizable Driving World Model with Video Mask Reconstruction

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Abstract

*World models that forecast environmental changes from actions are vital for autonomous driving models with strong generalization. The prevailing driving world model mainly build on video prediction model. Although these models can produce high-fidelity video sequences with advanced diffusion-based generator, they are constrained by their predictive duration and overall generalization capabilities. In this paper, we explore to solve this problem by combining generation loss with MAE-style feature-level context learning. In particular, we instantiate this target with three key design: (1) A more scalable Diffusion Transformer (DiT) structure trained with extra mask construction task. (2) we devise diffusion-related mask tokens to deal with the fuzzy relations between mask reconstruction and generative diffusion process. (3) we extend mask construction task to spatial-temporal domain by utilizing row-wise mask for shifted self-attention rather than masked self-attention in MAE. Then, we adopt a row-wise cross-view module to align with this mask design. Based on above improvement, we propose MaskGWM: a **Generalizable driving World Model** embodied with **Video Mask** reconstruction. Our model contains two variants: MaskGWM-long, focusing on long-horizon prediction, and MaskGWM-mview, dedicated to multi-view generation. Comprehensive experiments on standard benchmarks validate the effectiveness of the proposed method, which contain normal validation of Nuscene dataset, long-horizon rollout of OpenDV-2K dataset and zero-shot validation of Waymo dataset. Quantitative metrics on these datasets show our method notably improving state-of-the-art driving world model. Our project page is <https://github.com/SenseTime-FVG/OpenDWM>.*

1. Introduction

As an pivotal application of artificial intelligence, autonomous driving technologies, which require comprehending the surrounding environment and executing correct actions, have achieved significant advancements following the emergence of various learning models [19, 24] with scal-

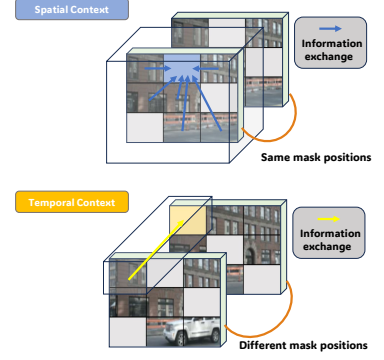
ability. However, the challenge of limited generalization to complex and varied scenarios remains unresolved for state-of-the-art methods [25]. For example, perception may encounter performance drops [44] in cases like weather changes, scene variations and motion blur. A promising solution for this problem is the use of world models, which directly predict environment changes under different actions. These models facilitate unraveling the complexities of data distributions and craft intricate regular patterns like human perception system [22].

Recently, advanced methods [12, 18, 39, 40, 42, 43] developed world models through diffusion-based generation task, capitalizing on the rapid development of advanced image generation systems [1, 7, 33]. Despite generating high-fidelity results, these approaches still struggle with long-horizon prediction and zero-shot generalization. To address this, GenAD [46] attempts to conduct training on large-scale OpenDV-2K [46] dataset with carefully-designed temporal modules, while VISTA [12] further introduces explicit re-weighted generation loss on structural and moving areas. However, two problems still exist in building a generalizable world model for autonomous driving. First, the combination of large-scale training dataset with more scalable transformer architectures is still under exploration. Second, one fundamental question remains unanswered: Is diffusion-based generation sufficient to build a generalizable world model? Since diffusion loss targets at iterative de-noising, the learning of visual semantics may not be straightforward. For example, MaskDiT [50] has shown diffusion models are complementary to well-known self-supervised methods [14], benefiting both convergence speed and generation quality.

Based on these insights into the diffusion pipeline and data for our driving world model, we design a new model, dubbed as MaskGWM, aiming at improving the fidelity, generalizability and long-time series prediction of the existing methods. Additionally, our model can also generate multi-view cases, by incorporating a multi-view module. We adopt Diffusion Transformer (DiT) as our backbone, which is more scalable and could take the information from a variety of datasets. Moreover, we introduce the mask

Method	Data scale	Model Setup		Traget
		Framework	Multi-view	
Drive-WM [40]	5h	Unet	✓	Diff
DiVE [20]	5h	DiT	✓	Diff
GenAD [46]	1740h	Unet	×	Diff
Vista [12]	1740h	Unet	×	Diff
MaskGWM(Ours)	1740h	DiT	✓	Diff+MR

(a) Real-world multi-view driving world models.



(b) Different context for mask reconstruction.

Figure 1. (a). MaskGWM improve fidelity and generalization from web-scale dataset, scalable DiT architecture and Mask Reconstruction (MR) target. (b) proposed MR apply a two branch structure for spatial context (scene objects) and temporal context (object motions)

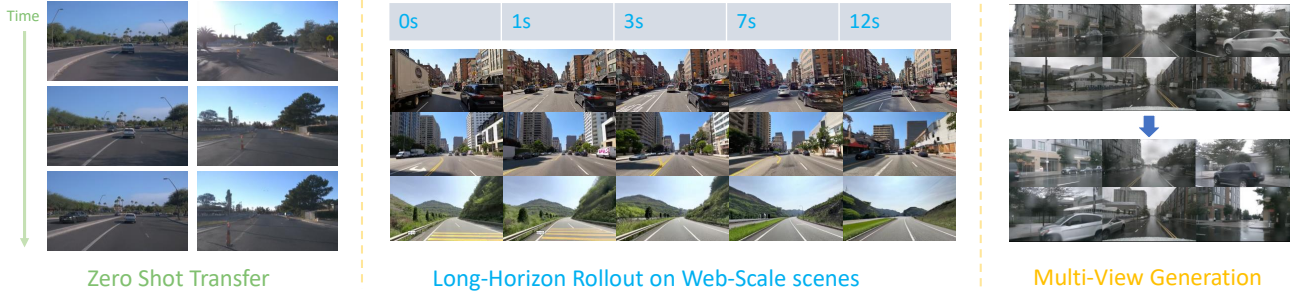


Figure 2. Our model facilitates zero-shot generation, consistent long-horizon prediction and multi-view video generation.

reconstruction as a complementary task for generation. Several impactful works [14, 45] have demonstrated that masked autoencoder is a powerful self-supervised method for representation learning from large-scale data and it is also extended to some diffusion methods [11, 50, 52] as an additional supervision to improve the models’ performance. Additionally, the features obtained by self-supervised learning is more contextually meaningful [5], which can be used as an auxiliary supervision to further improve the generation quality [18] However, integrating existing mask reconstruction for image generation into driving world models is not straightforward. There are still two questions to answer: (1) How can we enhance the synergy between diffusion model and mask reconstruction? Though the Mask reconstruction improves the contextual reasoning ability, this task contradicts with diffusion steps which include high noise ratio obscuring the feature details. (2) What kind of mask strategy should we use for video data? Different from image generation, the prediction of driving future requires an understanding of not only the objects within the scene but also their dynamic movements.

Therefore, we develop several special designs to address the aforementioned issues: (1) We make use of the mask tokens to improve the synergy between mask recon-

struction and diffusion models. Specifically, we propose a diffusion-related mask tokens (Sec.3.2) to initialized the invisible patches after DiT encoder. This special mask token can balance the learning of global and local features. (2) We design a novel two-branch mask reconstruction strategy. For spatial modeling, we use a mask shared across all frames and reconstruct invisible tokens via spatial transformer, this mask strategy is similar to some video mask modeling methods [36, 38]. For temporal modeling, we introduce a frame-specific mask and recover masked tokens via temporal transformer. Unlike the spatial branch, we directly link the unaligned tokens on temporal dimension after masking, which can be seen as a shift augmentation restricted by a new-proposed row-wise policy. We find this temporal branch achieves both masked patches prediction in the temporal context and a reduction in training costs.

In summary, our main contributions are:

- We propose MaskGWM, a generalizable DiT-based driving world model capable of forecasting long-term futures across web-scale scenes.
- We introduce mask reconstruction as a complementary task for diffusion-based world model.
- Comprehensive experiments on nuScene, OpenDV-2K and Waymo datasets demonstrate the superior video qual-

ity and robust generalization capabilities across extended time spans and different viewpoints.

2. Related Works

2.1. World Models

World models aim to infer the dynamic environment and ego state from past observations for accurate future predictions and planning. Most studies achieve world understanding by enabling the model to generate realistic videos that align with physical principles. In autonomous driving area, world models primarily focus on generating controllable real-world driving scenarios. GAIA-1 [18] employs an autoregressive transformer to predict tokens based on past state and then leverages a diffusion decoder to generate high-quality videos. In contrast, DriveDreamer [39] directly use diffusion model to represent complex environments and generate driving videos from multi-modal inputs. Drive-WM [40] extends to consistent and controllable multi-view video generation, exploring its application in end-to-end planning. GenAD [46] enhances generalization capability of world models by training on large-scale datasets and Vista [12] achieves further improvements by introducing attention on structure and dynamic area.

2.2. Diffusion Models with Self-supervised Learning

Recently, diffusion-based methods [13, 15, 32, 48] have become the mainstream of image and video generation. One important advancement in this field is Diffusion Transformer (DiT) [30]. Due to its better scalability and lower computational cost, DiT has been successfully applied in various diffusion models, achieving state-of-the-art results [7, 8, 20, 51].

On the other hand, masking strategies from self-supervised learning have been effectively applied to enhance generative models. With the development of DiT, research has focused on migrating this self-supervised approach to diffusion-based models. Initial works like MDT [11] modify DiT blocks to an asymmetric masking diffusion transformer architecture, where the encoder handles unmasked tokens only and a side-interpolator is introduced to recover the latent to the original shape. Following MDT [11], MaskDiT [50] simply utilizes a learnable token to fill in the masked places. SD-DiT [52], noticing the training-inference discrepancy and fuzzy relations between mask strategy and diffusion process, introduce a novel masking DiT with self-supervised discrimination. Despite their success, none of them applied the masking diffusion to the video generation models. Our work make this attempt on driving world model by applying different design of spatial context and temporal context, which focus on scene objects and nuance motion separately.

3. Method

Fig. 3 illustrates an overview of the proposed pipeline. MaskGWM builds upon Stable Diffusion 3 (SD3) [7], which is a well-studied DiT-based Text-to-Image (T2I) generation model, and introduce additional spatial and temporal blocks to extract the cross-view and cross-frame information. Moreover, we deploy a mask reconstruction module during training to improve the performance of our model. In this section, we first briefly review our DiT-based driving world model in Sec. 3.1. Then, Sec. 3.2 details the pipeline for mask reconstruction and introduces novel diffusion-related mask tokens designed to enhance the synergy between diffusion generation and mask reconstruction. Following this, Sec. 3.3 describes the extension of mask reconstruction to the temporal dimension. Finally, in Sec. 3.4, we describe the details of our cross-view module.

3.1. Preliminaries

Diffusion Models. A multi-view video sampled from a video dataset p_{data} can be represented as $x_0 \sim p_{data}$, where $x_0 \in \mathbb{R}^{T \times K \times C \times H \times W}$ is a sequence of T frames with view K , height H and width W . We first transform x_0 into video tokens $z_0 = P(\Theta(x_0)) \in \mathbb{R}^{T \times K \times C \times \hat{H} \times \hat{W}}$ via latent encoder Θ and patch encoding P . MaskGWM applies Rectified Flow [7, 26, 27] to model the generation process. Specifically, given a standard normal distribution $\epsilon \sim \mathcal{N}(0, I)$, Rectified Flow defines the intermediate noisy state as $z_\tau = (1 - \tau)z_0 + \tau\epsilon$, where $\tau \in [0, 1]$ is the diffusion timestep. The training target of Rectified Flow adopts the v -prediction, defined as:

$$\mathcal{L} = \mathbb{E}_{z_0, \epsilon \sim \mathcal{N}(0, I), \tau} [\|G_\theta(z_\tau, \tau, c, \mathcal{M}) - (z_0 - \epsilon)\|_2^2], \quad (1)$$

where c is the condition, $\mathcal{M}$ is a binary mask for mask reconstruction and G_θ is the DiT model.

Temporal Modeling. To facilitate temporal context learning, we attach a temporal transformer block after each 2D spatial transformer block, following common practice in video generation models [2, 29]. During the forward process, to standardize the inputs to the different self-attention layers in the transformer blocks, we reshape the video latent representation to $(TK)(\hat{H}\hat{W})C$ for spatial self-attention and to $(K\hat{H}\hat{W})TC$ for temporal self-attention. Additionally, we introduce reference frames according to video DiT models [9, 29]. During diffusion process, diffusion timestep τ of the reference frames is always set as 0, while the following frames to be predicted are embedded with regular timestep.

Unified Action Conditioning. Following VISTA, we provide nuanced control over low-level actions including angle, speed, trajectory and goal point, combining with high-level command capabilities. We construct action embedding by concatenating the Fourier embeddings [35] of all actions. Subsequently, these action embeddings are projected and

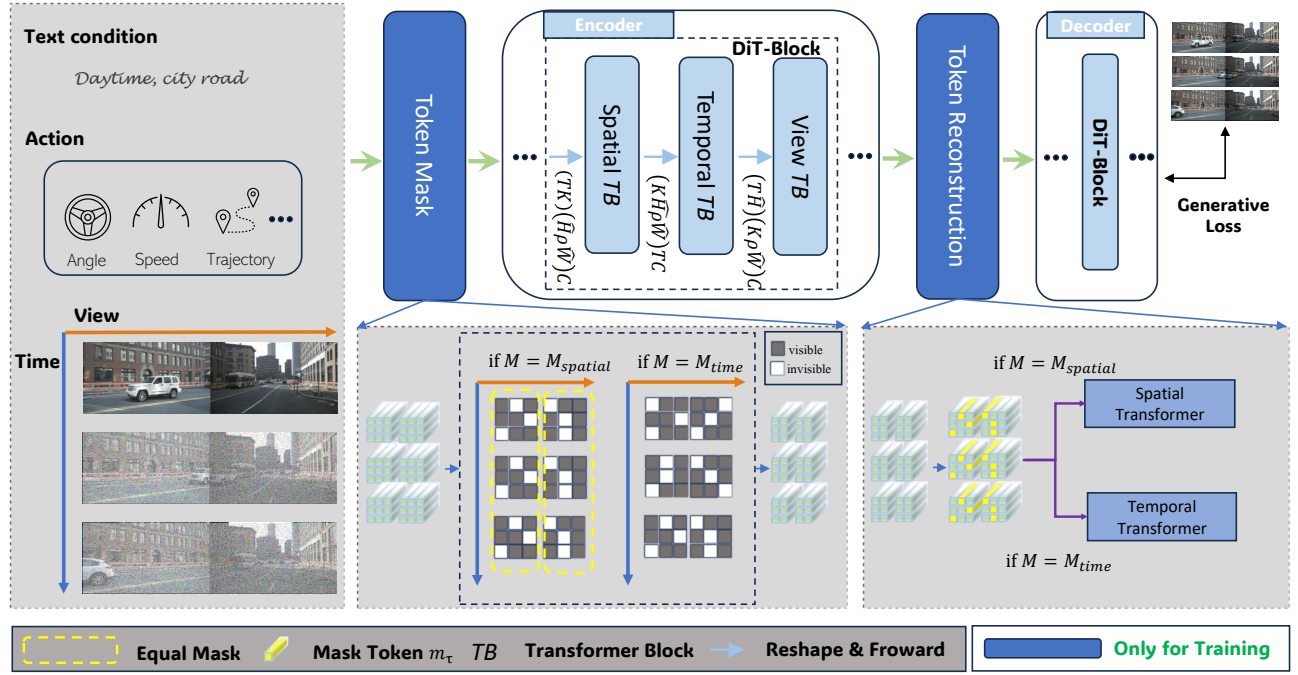


Figure 3. Overview of the MaskGWM. We propose mask reconstruction containing token mask and token reconstruction as a complementary task for training driving world model. **Token Mask**: we randomly sample tokens by temporal-shared $\mathcal{M}_{spatial}$ and temporal-unshared $\mathcal{M}_{time}$, specialized for spatial and temporal modeling. **Token Reconstruction**: we fill invisible tokens by diffusion-related mask tokens (Sec.3.2) and recover features by a two-branch transformer. Moreover, we introduce a row-wise mask strategy (Sec.3.3) for temporal branch. $\rho = 1 - r$ is used for simplicity in encoder.

added to the key and value features of the cross-attention layers in temporal transformer blocks.

3.2. Diffusion-related Mask Reconstruction

Motivated by previous works [50, 52], Masked Image Modeling (MIM) [14, 45] with mask reconstruction object has been adopted to diffusion-based generation model and achieve improvement on training efficiency and local contextual perception. However, these methods fail to consider the influence of diffusion process, which incorporate noise schedule and complex training target. Therefore, we propose this novel pipeline integrated with mask reconstruction and then introduce our diffusion-related mask tokens for compatibility with diffusion process, which can reduce the effect of noise during diffusion process.

Mask Reconstruction. During the training phase, the DiT backbone is asymmetrically divided into an encoder E and a decoder D for mask reconstruction and includes two additional processing steps. In the encoding stage, MaskGWM produces a token mask by random sampling a binary mask $\mathcal{M} \in \mathbb{R}^{T \times K \times 1 \times H \times W}$, given the video latent z_τ at timestep τ . Similar to [11, 52], $\mathcal{M}$ is only used to partition z_τ into visible patch tokens $z_\tau^v = z_\tau \odot \mathcal{M}$ and invisible

patch tokens $z_\tau^{iv} = z_\tau \odot (1 - \mathcal{M})$. In the decoding stage, an extra token reconstruction module is introduced to handle dropped invisible patches. Mask tokens m_τ , representing invisible patches, are infilled at the positions of the dropped tokens. Then, a transformer block F is utilized to provide contextual awareness from visible patches. In details,

$$G_\theta(z_\tau, \mathcal{M}) = D(F(E(z_\tau^v) \odot \mathcal{M} + m_\tau \odot (1 - \mathcal{M}))), \quad (2)$$

where τ, c in Eq.(1) are ignored for simplicity. Note that mask reconstruction is skipped for inference in which all tokens are visible, equivalent to $G_\theta(z_\tau) = D(E(z_\tau)) = D(E(z_\tau^v + z_\tau^{iv}))$. In practice, invisible patches z_τ^{iv} are directly dropped during the encoding of training to enable memory and speed benefits.

Diffusion-related Mask Tokens. SD-DiT [52] describes a fuzzy relationship between generation process and mask reconstruction. Concretely, mask reconstruction focuses on context reasoning while generation diffusion process aims to model the translations between real and fake distributions. From the viewpoint of diffusion model, the mask reconstruction for represent learning can be regarded as z_0 -prediction task, whereas rectified flow employs v -prediction ($\text{pred } z_0 - \epsilon$). Therefore, simply combining these two dis-

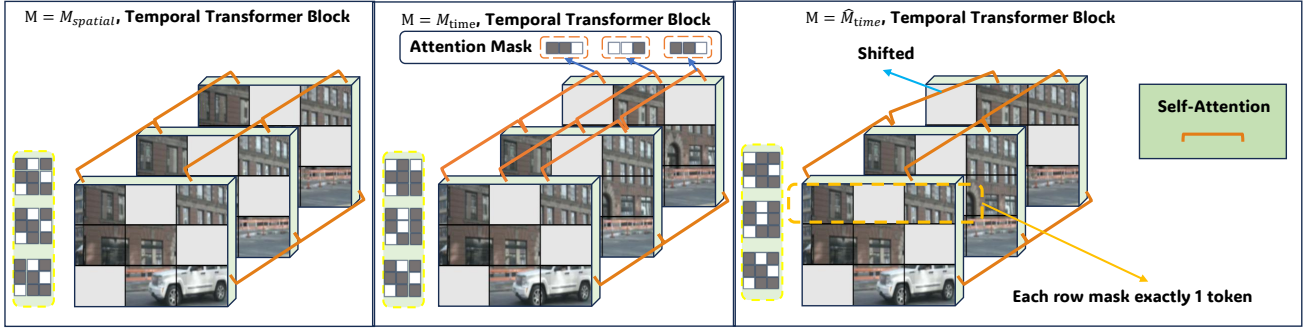


Figure 4. The comparison of different mask types and attention operations for temporal transformer block with mask reconstruction task. Attention mask is only applied when $\mathcal{M} = \mathcal{M}_{time}$

tinct objectives may not lead to good performance of the diffusion model, as also demonstrated by MaskDiT [50].

Existing works either instantiate mask tokens as learnable parameters [11, 14] p or directly take noisy tokens z_τ as input [52]. As discussed above, these mask tokens cannot balance these two targets due to the absence of explicit information for acquiring ϵ . We bridge this gap by introducing $f_m(\epsilon)$ into the mask token, where f_m is a small network for encoding noise ϵ . Given that ϵ is explicitly provided, it is easier to recover the original mask reconstruction target for representation learning within diffusion pipeline. Other than explicitly alignment for two prediction tasks, we further take τ into consideration. Overall, we define our mask token with learnable parameters p as:

$$m_\tau = (1 - \tau)f_m(\epsilon) + \tau p. \quad (3)$$

Based on our experiments in Section 4.4, we designed this mask token accordingly. When τ is large, the generation is performed at a high-noise level, and fine-grained image details are unknown. We hypothesize that the learnable parameters can estimate an average distribution under specific conditions (e.g., text) and assist in guiding the prediction direction when appearance details are lacking [47]. Conversely, on a low-noise level, mask reconstruction encourages the model to be attentive to local details of visible patches. Therefore, the model can leverage the visible information to recover the patches filled with noise (by $f_m(\epsilon)$).

3.3. Mask Reconstruction Strategy

Temporal and Spatial Mask. Previous methods [36, 38] extend MIM to video domain by sharing the random mask across time, where the mask $\mathcal{M} = \mathcal{M}_{spatial} = [\mathcal{M}^1, \mathcal{M}^2, \dots, \mathcal{M}^T]$ satisfy $\mathcal{M}^i = \mathcal{M}^j$ when $i \neq j$. Despite the effectiveness on understanding tasks in the aforementioned methods, this masking strategy may not be suitable for temporal modeling on driving video prediction. To in-

corporate temporal learning, we introduce a temporal unshared mask $\mathcal{M}_{time}$ satisfy $\mathcal{M}^i \neq \mathcal{M}^j$ when $i \neq j$. Then, we specialize $\mathcal{M}_{spatial}$ for spatial modeling and consider MIM with $\mathcal{M}_{time}$ for temporal modeling. Then, we make tasks synergy by devising a two-branch transformer block:

$$F = \begin{cases} F_s & \text{if } \mathcal{M} = \mathcal{M}_{spatial} \\ F_t & \text{if } \mathcal{M} = \mathcal{M}_{time} \end{cases}, \quad (4)$$

where $\mathcal{M} \in \{\mathcal{M}_{spatial}, \mathcal{M}_{time}\}$, F_s is a spatial transformer block and F_t is a temporal transformer block.

Row-wise Approximation. The most straightforward strategy for masking video data is to directly apply a random mask $\mathcal{M}_{time}$ on each frame like [36, 38], as shown in the middle column of Fig. 4. However, this design cause tokens on masked positions should be masked rather than directly dropped, since temporal self-attention require all entries have the same sequence length and apply an attention mask to control sparsity. As a result, this masked temporal self-attention requires a 3D attention mask to skip masked tokens, which leads to additional computational cost and precludes the direct application of optimization operators like FlashAttention [6]. To address the aforementioned issues, we employ shifted temporal self-attention for $\mathcal{M}_{time}$. As illustrated in the right column of Fig. 4, we randomly mask the same number of tokens for each row. Similar to the spatial branch, all invisible tokens are dropped, and the visible tokens are directly connected. Consequently, this operation can be regarded as a shift in temporal self-attention, which adheres to the core idea of Masked Reconstruction (MR): masked tokens are invisible and are predicted from context during the decoding stage. Additionally, rearranging the visible tokens row by row ensures the relevance of information in the temporal attention block during the encoding stage. In experiments, we find this design improves not only the training speed but also the generation metrics, especially on larger mask ratio r . Since this row-wise shifting allows

nearby tokens to fill in the blanks of masked tokens, We analysis this phenomenon by token filling makes all tokens on temporal axis are retained and minor shift can facilitate the temporal block in context reasoning.

To formulate this row-wise temporal mask, we define the mask ratio as r . To generate the mask for a $\hat{H} \times \hat{W}$ image latent at frame t , we randomly generate $\hat{H}$ one-dimensional mask, each having $r\hat{W}$ zero values, and concatenate them to formulate the final mask for this image, denoted by $\hat{\mathcal{M}}_{\text{time}}$. We use $\hat{\mathcal{M}}_{\text{time}}$ to replace $\mathcal{M}_{\text{time}}$ in our training.

3.4. Mask Reconstruction for Cross-View

Our method can be extended to generate multi-view driving videos by introducing view transformer blocks. We propose a cross-view row-wise self-attention mechanism, which concatenates video features horizontally and computes attention across multi-view features on the same row. Specifically, given a feature tensor of shape $TKC\hat{H}\hat{W}$, we apply masking, resulting in a tensor of shape $TKC\hat{H}[(1-r)\hat{W}]$. We then reshape it into size $(T\hat{H})[K(1-r)\hat{W}]C$.

The key insights of the proposed cross-view row-wise attention are twofold: First, since the vertical context can be modeled by spatial transformer blocks, row-wise feature exchange provides sufficient receptive field to extract multi-view information. Second, the proposed row-wise masking for reconstruction can also be utilized as data augmentation, since $Kr\hat{W}$ tokens of each row are randomly dropped. Because the proposed masked modeling task focuses on spatio-temporal modeling, we do not apply mask reconstruction along the view dimension.

4. Experiments

4.1. Setup

Datasets. We conduct comprehensive experiments on a single-view dataset OpenDV-2K [46] and two multi-views datasets nuScenes [3] and Waymo [34]. We follow the official splits to divide the training and validation sets.

Evaluation. The quality of the generated images and videos are assessed using the Frechet Inception Distance (FID) [16] for images and the Frechet Video Distance (FVD) [37] for videos. For fair comparison with previous works, we apply different settings for single-view model and multi-view model. For single view model, we align the evaluation setting of VISTA [12]. For multi-view model, we adopt the setting of Drive-WM [40]. Please refer to Appendix.7.3 for details. To assess generalization ability, we evaluate zero-shot performance on Waymo validation set, using 600 videos for FVD and 15K frames for FID.

4.2. Training Scheme

Our model is initialized with SD3 [7] medium checkpoint with 2B parameters. There are three stages for our training:

Method	Multi-view	Future-layout	FID↓	FVD↓
DrivingDiffusion [23]	✓	✓	15.8	332.0
Panacea [41]	✓	✓	16.9	139.0
MagicDrive [10]	✓	✓	19.1	218.1
DiVE [20]	✓	✓	-	94.6
DriveDreamer-2 [49]	✓	×	11.2	55.7
Drive-WM [40]	✓	×	15.8	122.7
MaskGWM-mview	✓	×	8.9	65.4
DriveGAN [21]	×	×	73.4	502.3
GenAD [46]	×	×	15.4	184.0
Vista [12]	×	×	6.9	89.4
MaskGWM-long	×	×	5.6	92.5

Table 1. Performance comparison with state-of-the-art methods on nuScene Dataset. The varying shades of gray indicate our multi-view metric following Drive-WM and single-view metric following Vista for more fair comparison. Future layout refers to the availability of layout information for future time steps.

Method	FVD↓	FID↓
VISTA [12] [†]	176.56	9.76
MaskGWM-long	118.83	9.55

Table 2. Zero-shot metrics on 600 Waymo validation samples. [†] denotes inference by official checkpoint.

Stage 1 for pre-training on large-scale OpenDV-2K dataset, Stage 2 for single-view model MaskGWM-long and Stage 3 for multi-view model MaskGWM-mview. For all stages, we resize the original images to 512×288 , masking ratio r is set to 0.25.

Stage 1. Following VISTA [12], we first pre-train our model on OpenDV-2K dataset. As our model starts from image backbone, we first train our model using single frame videos with batch size 768 for 18K iterations and then we train our temporal blocks with 16/20/24 frame videos and batch size 64. In particular, temporal blocks is initialized with zero and the training of temporal blocks takes 24K iterations in this step. Afterwards, we insert the zero-initialized reconstruction blocks F and train with masking strategy for extra 20K iterations. The reason for varying frame length during training is that the frame numbers for single-view and multi-view models are different.

Stage 2 for MaskGWM-long. We devised MaskGWM-long to align with VISTA, where the frame length T is set to 25 and cross-view blocks are skipped. We also follow the stage 2 of VISTA’s collaborative training, in which data were sampled equally from nuScene and OpenDV-2K, and the action module is zero-initialized and trained.

Stage 3 for MaskGWM-mview. Based on MaskGWM-



Figure 5. **Long-horizon prediction results of MaskGWM.** Our model is capable of forecasting long video sequences with stability, devoid of collapse or blurring issues.

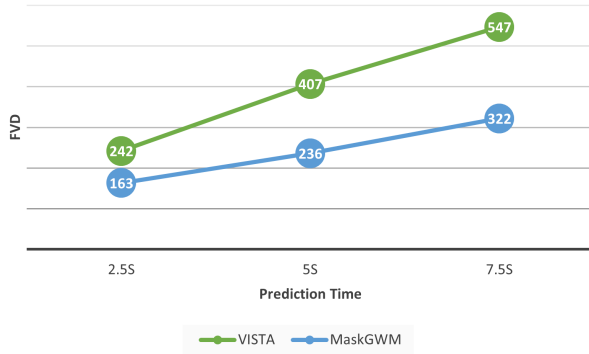


Figure 6. **Comparison of Long-horizon FVD metric on OpenDV2K validation set.** Our method demonstrates superior performance in terms of both value and growth rate.

long, we enable multi-view ability of MaskGWM-mview by adding the cross-view blocks and train it on nuScene dataset for 6K steps and T is reduced to 8.

4.3. Comparison

Generation Quality. Tab.1 presents the quantitative comparison. In single-view generation, we achieve a remarkable FID of 5.6 and an FVD of 92.5, surpassing the previous state-of-the-art approaches [12, 46] that are also trained on web-scale dataset. For multi-view cases, similar improvement can also be observed, with 8.9 FID and 65.4 FVD. It is worth noting that our model directly predict multi-view future without the requirement of future layouts, unlike those methods [20] deviating from video prediction task. Furthermore, our method represents a pioneering effort in extending a generalizable single-view model, trained on OpenDV-2K, to the domain of multi-view models.

Generalization ability. In Tab.2, we assess the generalization capability of our approach on the Waymo dataset, which is excluded from our training datasets. We conducted

inference using the official checkpoint of VISTA [12]. The results indicate that our method attains superior FVD while maintaining comparable FID, thereby demonstrating the generalization ability of our method.

Long-horizon prediction. Fig.6 illustrates the comparison of long-time prediction with VISTA. Due to the computational cost of generating long videos, both FID and FVD metrics are calculated on 300 videos randomly sampled from the OpenDV-2K validation set. The slope of FVD curve is significantly lower than VISTA, showing the less degradation. Qualitative results are illustrated in Fig.5.

4.4. Ablation Studies

We conduct comprehensive ablation study to verify the performance of every component in MaskGWM. Following the setting of GenAD for effective experiments, most ablation studies are conducted on stage 1 after mask reconstruction is inserted and the metrics are reported on the validation set of OpenDV-2K dataset with 3000 video clips for FVD and 18000 frames for FID. For the ablation of cross-view transformer blocks, we use multi-view nuScene metrics.

Effect of mask tokens. As indicated in Tab.3a, we make comparisons across different designs for mask tokens m_τ . To analysis the influence of diffusion timestep τ , we test only applying mask reconstruction on certain timestep range. We find learnable parameters p shown better results on high noisy level and proposed noise embedding $f_m(\epsilon)$ perform better on low noisy level. This demonstrates the merit of noise embedding, which can recover the original mask reconstruction target for represent learning by explicitly giving ϵ , achieving better result on generation steps with local details. In addition, we also try the mask tokens used in SD-DiT [52], combining with extra contrastive [4] loss. The experiment shows our proposed diffusion-related mask tokens achieve best performance via combining the advantage of p and $f_m(\epsilon)$.

Effect of mask reconstruction. In Tab.3b and Tab.4a, we

m_τ	τ range	contra.	FVD↓	FID↓
p	[0, 1]		126.71	7.35
p	[0.5, 1]		120.35	7.12
$f_m(\epsilon)$	[0, 1]		116.85	6.40
$f_m(\epsilon)$	[0, 0.5]		109.87	5.92
z_τ	[0, 1]	✓	113.26	6.32
ours	[0, 1]		105.52	5.69

(a) Different design of m_τ .

$\mathcal{M}_{time}$	row	shift att.	$\mathcal{M}_{spatial}$	r	FVD↓	FID↓
				0	136.55	10.28
✓				0.25	142.68	10.75
✓	✓			0.25	143.07	10.39
✓	✓	✓		0.25	121.38	7.36
			✓	0.25	116.73	5.92
✓	✓	✓	✓	0.25	105.52	5.69

(b) Ablations of two-branch mask reconstruction.

Table 3. Ablations of the components in our mask reconstruction. “contra” stands for contrastive loss, “row” denotes whether $\mathcal{M}_{time}$ satisfies above row-wise generation strategy ($\mathcal{M}_{time} = \hat{\mathcal{M}}_{time}$) and “att.” refers to self-attention

row&shift att	r	FVD↓	Time↓
	0.1	133.24	0.368d
	0.25	142.68	0.352d
✓	0.1	123.70	0.357d
✓	0.25	121.38	0.329d

(a) Different mask ratio r .

two-branch	layers	FVD↓	FID↓
	1	121.37	5.85
✓	1	105.52	5.69
	2	127.91	6.03
✓	2	107.34	5.60

(b) Different design of F .

att dim	ratio r	FVD↓	FID↓
KW	0	65.9	9.2
KW	0.25	65.4	8.9
K	0.25	71.5	8.7
KHW	0.25	64.7	8.5

(c) Ablation of cross-view block.

Table 4. Ablations of the impact of mask ratio r , different view transformer block designs and the effect of mask reconstruction on convergence. “share” stands for applying one shared model for F_s and F_t . **Time** indicate the training time for 10k steps.

explore the effectiveness of proposed mask reconstruction in spatial-temporal domain. According to Tab.4a, we find that our row-wise time mask with shift attention mechanism assists a lot the model’s convergence, while larger mask ratio can also make positive effect on the results. Thus, this setting is adopted in the ablation study on the mask strategy $\mathcal{M}$. As shown in Tab.3b, combining $\mathcal{M}_{spatial}$ and $\mathcal{M}_{time}$ can significantly improve the generation quality of the whole videos and each single images. We hypothesis this is because the temporal modeling is more sensitive to the dropout ratio than spatial counterpart (Please see more details on Appendix.6.1). When shifted temporal self-attention is applied with a row-wise mask, all tokens on temporal axis are retained and experience only minor spatial shifts. Whereas invisible tokens are skipped without shift temporal self-attention, which results in a discrepancy between the number of tokens used in training and inference. Moreover, the training speed is improved since invisible tokens dropped. From Tab.3b, we can see our two-branch mask reconstruction achieve best results, showing the effectiveness on temporal context.

Effect of cross-view module. As shown in Tab.4c, we make comparisons across different types of cross-view modeling. We find that introducing mask reconstruction on stage-3 training also yields favorable results. This indicates that randomly masking certain tokens along the view-row dimension is not detrimental and can even enhance the final results. This is different from temporal counterpart that requires shift-attention to avoiding tokens dropping.

Furthermore, the experiment shows that proposed atten-

tion on dimensions KW outperforms view-attention on K used in previous methods [39, 40]. For view attention on KHW, despite the minor improvement, the computation complexity explodes significantly. Consequently, this design is not adopted in our experiments.

Effect of two-branch token reconstruction. We validate the effectiveness of two-branch transformer reconstruction structure for token reconstruction on Tab.4b. We remove two-branch structure by applying sequential spatial-temporal transformer blocks, which is shared by F_s and F_t . We find the two-branch structure produces better results, especially on FVD. Unshared design forces the model to reconstruct masked features by corresponding context, leading to better spatial and temporal modeling.

5. Conclusion

We introduce MaskGWM, a DiT-based driving world model trained on web-scale datasets with masking strategy during training. By introducing a novel video dual-branch mask reconstruction, our model excels in both numeric metrics on fidelity and generation ability. Additionally, our mask policy also accelerate the training process and reduce memory consumption. Our extensive experiments show MaskGWM achieves the state-of-the-art performance on generation quality on nuScene, zero-shot ability on Waymo and long-time prediction ability on OpenDV-2K. These results further indicates that our method can serve as a training programs to enable long driving video prediction.

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